

Administrator Guide

"WEATHERLAB"

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ADMINISTRATOR GUIDE

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1. About This Guide

This guide is intended for everyone who looks after WeatherLab, i.e. the project team and the administrators responsible for the station. It explains how to connect to the Raspberry Pi, transfer files, check services, change settings, install updates and troubleshoot problems. Day-to-day use of the display is covered in the User Manual, and the technical design in the Technical Documentation.

1.1 WHAT YOU NEED

WHAT	PURPOSE
Windows PC or laptop	used to manage the station
PuTTY	runs commands on the Raspberry Pi (free, putty.org)
FileZilla	copies files between the PC and the Raspberry Pi (free, filezilla-project.org)
Login credentials	user name and password of the Raspberry Pi, available from the project team
Network	the PC must be on the school network or on the station's Wi-Fi

1.2 ADDRESSES AND CREDENTIALS

ITEM	VALUE
User name	`weatherlabadmin`
Password	deliberately not included in this guide, ask the project team
Address on the school network	`weatherlab.local` or the IP address of the Pi (section 2.1)
Address on the station's Wi-Fi	`192.168.10.1`
Website	http://weatherlab.lan/ or http://192.168.10.1/
Project folder	`/home/weatherlabadmin/pi-dashboard`, or `~/pi-dashboard` for short

Passwords, the settings PIN and the Telegram bot token must never appear in documents, chats or screenshots (section 8.3).

1.3 IMPORTANT FILES

FILE	CONTENTS
`~/pi-dashboard/config.py`	settings: API key, LED pins, thresholds, times, bot token, PIN
`~/pi-dashboard/telegram_chats.json`	approved Telegram recipients and pending requests

FILE	CONTENTS
`~/pi-dashboard/locations.json`	saved locations and the active location
`~/pi-dashboard/history.db`	measurement history of the last few days
`/etc/hostapd/hostapd.conf`	the station's Wi-Fi: network name and password
`~/.config/labwc/autostart`	starts the browser in full screen on the display

2. Connecting with PuTTY

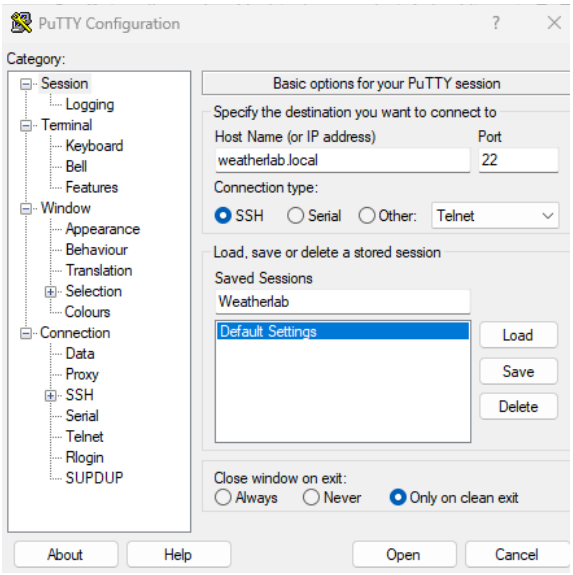
PuTTY opens a window in which you run commands directly on the Raspberry Pi, just as if you were sitting in front of it. The connection uses SSH and is encrypted.

2.1 CHOOSING THE RIGHT ADDRESS

- **On the school network:** enter `weatherlab.local`. If that does not work, use the IP address instead. You can find it in the router or display it on the Pi with `hostname -I`. It may change after a restart.
- **On the station's Wi-Fi:** connect to the WeatherLab Wi-Fi and enter `192.168.10.1`. This address never changes.

2.2 SETTING UP PUTTY

1. Start PuTTY. The **PuTTY Configuration** window opens.
2. Under **Host Name (or IP address)**, enter the address from section 2.1.
3. Leave **Port** at 22 and **Connection type** at **SSH**.
4. To avoid typing this every time, enter a name such as `WeatherLab` under **Saved Sessions** and click **Save**. Next time, simply double-click the name.
5. Click **Open**.



2.3 LOGGING IN

1. The very first time you connect, a **PuTTY Security Alert** appears, asking whether you trust this device. Confirm with **Accept**. The message will not appear again.
2. At `login as:`, enter the user name `weatherlabadmin` and press Enter.
3. Enter the password and press Enter. **No** characters or asterisks appear while you type. This is normal.
4. You are logged in when `weatherlabadmin@weatherlab:~ $` appears in green.

If **Access denied** appears instead, the password was wrong. Simply try again.

2.4 WORKING IN THE TERMINAL

The usual Windows keyboard shortcuts do not work in PuTTY. These are the important ones:

ACTION	HOW TO DO IT
Paste	**right-click** in the window, Ctrl+V does not work
Copy	select the text with the mouse, it is copied automatically
Cancel a command	**Ctrl+C**, for example to stop a running log
Repeat the last command	up arrow, then Enter
Complete a file or folder name	type the first few letters and press **Tab**
Clear the screen	`clear`
Log out	`exit` or close the window

Commands starting with `sudo` run with administrator rights. The first time you use `sudo`, the Pi may ask for the password again. It is also entered invisibly.

2.5 PASTING LONG COMMANDS AND SCRIPTS

Many changes come as a block of text to copy, for example the extension scripts supplied as text files. To paste them reliably:

1. Open the text on the PC, select everything with **Ctrl+A** and copy it with **Ctrl+C**.
2. **Right-click** once in PuTTY. With long texts this takes a few seconds.
3. Wait until the command prompt appears again. At the end, the script files print **SKRIPT IST DA** (“script is in place”).
4. Always enter commands that ask questions afterwards, such as … `installieren`, **separately** and not in the same block. Otherwise the script takes the remaining text as its answers.

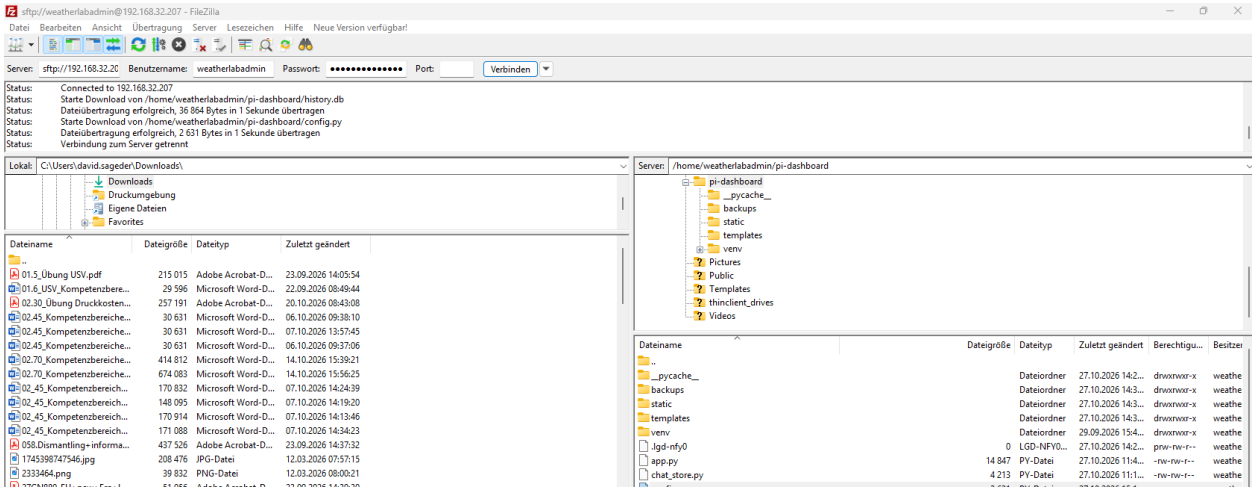
If a block contains exclamation marks, the message `event not found` may appear. In that case, enter `set +H` first. Our script files already include this in the first line.

3. Transferring Files with FileZilla

FileZilla shows the folders on the PC on the left and the folders on the Raspberry Pi on the right. To transfer files, simply drag them from one side to the other.

3.1 CONNECTING

1. Start FileZilla.
2. Fill in the bar at the top: **Server** `sftp://` followed by the address, for example `sftp://weatherlab.local`, **Benutzername** (user name) `weatherlabadmin`, the **Passwort** (password) and **Port** `22`.
3. Click **Verbinden** (Connect). The first time, confirm the message about the unknown host key with **OK**.
4. The folder `/home/weatherlabadmin` appears on the right. The project folder is `pi-dashboard`.



3.2 UPLOADING AND DOWNLOADING

- **To the Pi:** drag the file from left to right into the `pi-dashboard` folder.
- **To the PC:** drag the file or folder from right to left.
- **Check file names:** browsers sometimes add a number when downloading, for example `weatherlab-standorte_239.sh`. In that case, right-click the file on the right, choose **Umbenennen** (Rename) and restore the exact name.
- **Transfer queue:** FileZilla lists pending transfers at the bottom. Remove old entries with a right-click so that an old file is not uploaded by mistake.

System files under `/etc` cannot be overwritten with FileZilla because this requires administrator rights. They are edited in PuTTY with `sudo nano` (section 4.2).

4. Important Commands

4.1 BASIC COMMANDS

COMMAND	EFFECT
`cd ~/pi-dashboard`	change to the project folder
`ls -l`	list the files in the current folder
`cat file`	show the contents of a file
`nano file`	edit a file (section 4.2)
`cp file file.bak`	make a copy of a file
`mv old new`	rename or move a file
`hostname -I`	show the IP addresses of the Pi
`sudo reboot`	restart the Raspberry Pi
`sudo poweroff`	shut down the Raspberry Pi

4.2 EDITING FILES WITH NANO

1. Open the file, for example `nano ~/pi-dashboard/config.py`. For system files, put `sudo` in front.
2. Use the arrow keys to move to the right place and change the value. The mouse does not work here.
3. Save with **Ctrl+S**. Alternatively, press Ctrl+O and then Enter.
4. Exit with **Ctrl+X**.

Before making larger changes, it is worth making a copy, for example `cp config.py config.py.bak`. In config.py, text goes in quotation marks, numbers do not, and the indentation must not be changed.

4.3 RESTARTING AND SHUTTING DOWN

```
sudo reboot      # restart, everything is back after about a minute
sudo poweroff    # shut down, the plug can then be pulled
```

In everyday use, the power supply can simply be unplugged. If you are connected via PuTTY anyway, it is better to shut the Pi down first, as this makes sure all data has been saved.

5. Monitoring the Services

5.1 THE STATION'S SERVICES

Everything the station does runs as a service and starts automatically at power-on:

SERVICE	TASK
hostapd	broadcasts the station's Wi-Fi
dnsmasq	assigns IP addresses on the Wi-Fi and resolves weatherlab.lan
startup-wifi	sets the fixed address 192.168.10.1 at startup and starts the two services above
pi-dashboard	the application with sensor, weather, database and settings
apache2	serves the website on port 80
telegram-bot	the Telegram bot with alerts and the daily report

5.2 CHECKING STATUS AND RESTARTING

Check all services at once:

```
systemctl status hostapd dnsmasq startup-wifi pi-dashboard apache2 telegram-bot --no-pager |
grep -E "●|Active"
```

Every service should report **active (running)**, startup-wifi **active (exited)**. If **failed** appears anywhere, restarting the service is often enough:

```
sudo systemctl restart pi-dashboard     # website, sensor, settings
sudo systemctl restart telegram-bot     # Telegram bot
sudo systemctl restart hostapd          # station Wi-Fi
```

5.3 READING THE LOGS

Every service logs what it is doing. When there is a problem, the log almost always shows the cause:

```
sudo journalctl -u pi-dashboard -n 30 --no-pager    # the last 30 lines
sudo journalctl -u telegram-bot -f                  # follow live, stop with Ctrl+C
```

The bot token is automatically replaced with *** in the log, so a log can safely be sent to the project team for troubleshooting.

5.4 RASPBERRY PI HEALTH

```
df -h /                     # free disk space
free -h                     # memory
vcgencmd measure_temp       # CPU temperature
vcgencmd get_throttled      # power supply and overheating
uptime                      # time since the last start
```

For `get_throttled`, `throttled=0x0` means everything is fine. Any other value indicates that the voltage was too low or the Pi got too hot. If the message **Low voltage warning** appears on the display, the power supply is not delivering enough power. The Raspberry Pi 5 should be used with the official 27 W power supply.

5.5 QUERYING READINGS IN THE TERMINAL

The website fetches its values through small interfaces, which can also be queried directly in PuTTY. This shows whether the sensor and the weather query are working, even if the website is not displaying correctly at the moment:

```
curl -s http://localhost/api/indoor      # indoor values from the sensor
curl -s http://localhost/api/outdoor     # outdoor weather from OpenWeather
curl -s http://localhost/api/location    # active location
```

The responses are in JSON format. If `"temperature"` contains a number, that part is working. If it shows `null`, the sensor or OpenWeather is not currently delivering values.

6. Routine Tasks

6.1 WHAT CAN BE DONE ON THE DISPLAY

Many tasks no longer require PuTTY. In the settings tab behind the gear icon, you can use the PIN to approve, rename and remove Telegram recipients, change the Wi-Fi name and password, manage locations and change the PIN itself. Chapter 9 of the User Manual explains how. This chapter describes what can only be done via PuTTY.

6.2 FORGOTTEN PIN

The PIN is stored in config.py. This is how to display it or set a new one:

```
grep SETTINGS_PIN ~/pi-dashboard/config.py    # show the PIN
nano ~/pi-dashboard/config.py                  # change the line SETTINGS_PIN = "…"
sudo systemctl restart pi-dashboard            # apply the new PIN
```

The PIN must have 4 to 8 digits and is written in quotation marks. After 5 wrong attempts on the display, input is blocked for 2 minutes. Restarting pi-dashboard lifts the block.

6.3 CHANGING THE WI-FI MANUALLY

If changing it in the settings tab does not work, it can also be done directly in the configuration:

1. Open the file: `sudo nano /etc/hostapd/hostapd.conf`
2. Change the name in the `ssid=` line and the password in the `wpa_passphrase=` line. The password needs 8 to 63 characters and must not contain umlauts.
3. Save with Ctrl+S, exit with Ctrl+X.
4. Restart the Wi-Fi: `sudo systemctl restart hostapd`

Afterwards, all devices must reconnect using the new name or password. If you are connected with PuTTY via the station's Wi-Fi yourself, you will lose the connection at this moment. It is therefore best to make such changes over the school network. If hostapd finds an error in the file, `sudo hostapd -dd /etc/hostapd/hostapd.conf` shows the faulty line.

6.4 CHANGING THRESHOLDS AND TIMES

All values are stored in `~/pi-dashboard/config.py`. After a change, the corresponding service must be restarted:

ENTRY	MEANING	RESTART AFTERWARDS
`NIGHT_START`, `NIGHT_END`	night mode of the display (hours)	pi-dashboard
`IDLE_RETURN_SECONDS`	return to the home page after this many seconds	pi-dashboard
`GAS_QUALITY_THRESHOLDS`	limits for the air quality levels	pi-dashboard
`ALERT_TEMP_HIGH`, `ALERT_TEMP_LOW`	alert when too warm or too cold (°C)	telegram-bot
`ALERT_HUMIDITY_HIGH`, `_LOW`	alert for humidity (%)	telegram-bot
`ALERT_WIND_KMH`, `ALERT_FROST_C`	alert for storms and frost	telegram-bot
`ALERT_RAIN`	rain and snow notifications on or off	telegram-bot
`TELEGRAM_DAILY_REPORT`	time of the daily report, "" turns it off	telegram-bot
`TELEGRAM_QUIET_START`, `_END`	period in which alerts arrive without sound	telegram-bot

```
sudo systemctl restart pi-dashboard
sudo systemctl restart telegram-bot
```

6.5 RENEWING THE BOT TOKEN

If the bot token has accidentally become public, for example in a screenshot, it must be renewed:

1. In Telegram, open the chat with **@BotFather**, send `/revoke` and select the bot. BotFather sends a new token.
2. Open `nano ~/pi-dashboard/config.py` and enter the new token for `TELEGRAM_BOT_TOKEN`.
3. Restart both services, because the settings tab also needs the token to send test messages: `sudo systemctl restart telegram-bot pi-dashboard`

The old token stops working. The recipients are kept, as they are stored in telegram_chats.json.

6.6 CHANGING THE USER PASSWORD

The password for PuTTY and FileZilla is changed with `passwd`. You are asked for the old password first and then twice for the new one, all typed invisibly. Please keep the new password in a safe place. Without it, the system can only be accessed directly on the Pi with a keyboard and monitor.

6.7 ENTERING THE API KEY AND LED PINS

The OpenWeather key and the connections of the three status LEDs are also stored in `~/pi-dashboard/config.py`:

```
OPENWEATHER_API_KEY = "..."   # key from openweathermap.org, 32 characters
LED_ENABLED = True            # False turns the LEDs off
LED_PIN_GRUEN = 17            # GPIO 17 = Pin 11
LED_PIN_GELB = 27             # GPIO 27 = Pin 13
LED_PIN_ROT = 22              # GPIO 22 = Pin 15
```

Entering or replacing the API key:

1. Log in at openweathermap.org and copy the key under **My API keys**.
2. In PuTTY, open `nano ~/pi-dashboard/config.py` and paste the key between the quotation marks of `OPENWEATHER_API_KEY` (right-click to paste in PuTTY).
3. Save with Ctrl+S, exit with Ctrl+X.
4. Run `sudo systemctl restart pi-dashboard`.

A newly created key can be rejected for up to two hours. During this time, the log shows **Invalid API key** and the outdoor values stay empty. The same key is also used for the weather maps and the location search in the settings. The Telegram bot gets its values from the website and does not need to be restarted.

Changing the LEDs: config.py contains the GPIO numbers, not the pin numbers on the header. If an LED is moved to a different pin, change the corresponding number and then run `sudo systemctl restart pi-dashboard`. The LED wiring is described in section 5.2 of the Technical Documentation. The log shows whether the LEDs were detected:

```
sudo journalctl -u pi-dashboard -n 100 --no-pager | grep -i led
```

6.8 DATE AND TIME

Although the Raspberry Pi 5 has a built-in clock, it does not keep running without an additional battery when the power is off. The Pi therefore gets the time from the internet after starting up. If the time is wrong, the daily report and night mode will also run at the wrong time.

```
timedatectl                                  # show time, time zone and sync status
sudo timedatectl set-timezone Europe/Vienna  # set the time zone
```

System clock synchronized must show **yes**. If it shows no, the internet connection via the LAN cable is usually missing.

7. Updates and Maintenance

7.1 UPDATING THE SYSTEM

The operating system packages should be updated about once a month:

```
sudo apt update
sudo apt full-upgrade -y
sudo reboot
```

This takes a few minutes. After the restart, use the command from section 5.2 to check that all services are running.

7.2 EXTENSION SCRIPTS

New features were added to the station with two scripts. Both are located in the project folder:

SCRIPT	ADDS
`weatherlab-erweiterung.sh`	settings tab, on-screen keyboard, new Telegram bot
`weatherlab-standorte.sh`	locations, location picker, PIN change, renaming recipients

A new version is installed as follows:

1. Copy the file into the `pi-dashboard` folder with FileZilla, or use the text file for pasting into PuTTY (section 2.5).
2. Run it in PuTTY without sudo: `bash ~/pi-dashboard/weatherlab-standorte.sh installieren`
3. Answer the questions. Green lines with a tick mean that the step was successful.
4. At the end, confirm the restart with `j` (yes) and press **Ctrl+F5** in the browser on the PC.

7.3 REGULAR MAINTENANCE

To keep the station running reliably in the long term, a quick check according to this schedule is worthwhile:

WHEN	WHAT	HOW
weekly	look at the display: are the indoor and outdoor values current, is an LED lit?	time stamp boxes on the home page
monthly	check the services	section 5.2
monthly	check disk space, temperature and power supply	section 5.4
monthly	update the system	section 7.1

WHEN	WHAT	HOW
monthly	remove dust from the sensor and ventilation slots	dry brush, switch the Pi off first
as needed	remove Telegram recipients who no longer need messages	settings tab on the display
at the end of the project	change the password, PIN and Wi-Fi password, renew the bot token if necessary	sections 6.5 and 6.6

8. Troubleshooting

8.1 PROCEDURE

1. Connect with PuTTY and check all services (section 5.2).
2. For the service marked **failed**, read the log (section 5.3). The last error message usually names the file and line.
3. Restart the service.
4. If that does not help, restart the whole Pi with `sudo reboot`.
5. If it still does not work, inform the project team and include the error message.

8.2 COMMON PROBLEMS

PROBLEM	CHECK	SOLUTION
PuTTY: Connection timed out	is the PC on the same network, is the address correct?	check the address on the Pi with `hostname -I`, or connect via the station's Wi-Fi using 192.168.10.1
PuTTY: Access denied	user name and password	enter them again, paying attention to upper and lower case
Website: 503 Service Unavailable	`systemctl status pi-dashboard`	read the log, fix the error in config.py, restart the service
Outdoor values missing	`ping -c 3 api.openweathermap.org`	check the LAN cable, check the API key in config.py (section 6.7)
LEDs do not light up	`journalctl -u pi-dashboard | grep -i led`	check the wiring and the GPIO numbers in config.py (section 6.7); the sensor needs a few minutes after startup
Indoor values show only dashes	`i2cdetect -y 1` must show 76 or 77	check the sensor wiring, then restart pi-dashboard
Station Wi-Fi not visible	`systemctl status hostapd startup-wifi`	`sudo systemctl restart startup-wifi`, look for errors in hostapd.conf with `hostapd -dd`

PROBLEM	CHECK	SOLUTION
Bot does not respond	`sudo journalctl -u telegram-bot -n 30`	check the internet connection and token, `sudo systemctl restart telegram-bot`
Display only shows the desktop	is Chromium running?	`sudo reboot`, check the autostart file
New feature not visible on the PC	old version in the browser cache	Ctrl+F5
Low voltage warning	`vcgencmd get_throttled`	use the official 27 W power supply
Disk full	`df -h /` shows 100 %	delete files that are no longer needed, `sudo apt clean` empties the package cache
Script reports $'\r': command not found	the file was edited on Windows	`sed -i 's/\r$//' file.sh`, our scripts do this themselves

8.3 SECURITY NOTES

- Only share the Pi password, the Wi-Fi password, the PIN and the bot token with people who look after the station.
- Never show the bot token in screenshots. If it has become visible, renew it immediately (section 6.5).
- At the end of the project, change the user password (section 6.6) and set a new PIN.
- Do not run commands from the internet with sudo without checking them first.
- The website can only be reached on the school network and the station’s Wi-Fi, not from the internet, and this should stay that way.